Experiment 9: Water Jug Problem

Aim

To solve the Water Jug problem by generating a sequence of operations to measure an exact quantity of water using two jugs of different capacities.

Algorithm:

1. Represent state as (amount in jug1, amount in jug2).
2. Possible operations: Fill1, Fill2, Empty1, Empty2, Pour1to2, Pour2to1.
3. Use BFS to find the shortest sequence of operations to reach the goal amount.
4. Avoid revisiting states.

Code:

```
from collections import deque

def water_jug(cap1, cap2, goal):
    start = (0, 0)
    q = deque([(start, [])])
    visited = {start}
    while q:
        (j1, j2), path = q.popleft()
        if j1 == goal or j2 == goal or j1 + j2 == goal:
            return path + [(j1, j2)]
        states = [
            (cap1, j2), (j1, cap2), (0, j2), (j1, 0),
            (j1 - min(j1, cap2 - j2), j2 + min(j1, cap2 - j2)),
            (j1 + min(j2, cap1 - j1), j2 - min(j2, cap1 - j1))
        ]
        for s in states:
            if s not in visited:
                visited.add(s)
                q.append((s, path + [(j1, j2)]))
    return None

print("Solution:", water_jug(4, 3, 2))
```

Sample Output:

Output

```
Solution: [(0, 0), (0, 3), (3, 0), (3, 3), (4, 2)]
```

Result:

The algorithm found a valid sequence of operations to measure exactly 2 litres using 4L and 3L jugs.